Ramanujan, Landen, and Baker N-Series Refinement: Higher-Degree Modular Gap Laws and a Two-Axis Comparison

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Abstract

We compare two refinement mechanisms that arise in elliptic perimeter approximation and geometric acceleration. Ramanujan's second approximation to the ellipse perimeter acts along the eccentricity axis and reproduces the Gauss–Kummer expansion through order h^4. The geometric N-series acts along a resolution axis and cancels the even-power hierarchy in N^{-2}. These mechanisms are distinct on the ellipse, but they meet in the modulus variable of the Gauss–Legendre arithmetic–geometric mean iteration, where the normalized gap obeys the descending Landen contraction $\delta_{n+1} \sim \frac{1}{4}\delta_n^2$.

The leading higher-degree gap law $\delta(q^m)/\delta(q)^m \to 4^{1-m}$ is an immediate corollary of the classical expansion $\delta(q) \sim 4q$; we record it for completeness. The contributions of this note are three. First, for every fixed integer $m \geq 2$, we prove the first-correction law

$$\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\big(1 + 4m\,q^2 + O(q^4)\big), \qquad q \to 0.$$

Second, we combine the classical AGM error law with the Landen contraction to obtain the linkage constant $\alpha_n/\delta_n^2 \to 1/8 = 2/4^2$. Third, we show that this is consistent with the independent dynamic coefficient law $\alpha_{n+1}/\alpha_n^2 \to 1/2$.

The result does not identify fixed N-series extrapolation with modular transformation. It isolates a structural bridge: both frameworks select coefficients that move the active error variable into a higher-power regime. A companion Baker 4:3 trisection cascade supplies a separate geometric comparison point, where a normalized signed error satisfies

$$\eta_{k+1} = -\eta_k^3 + O(\eta_k^5).$$

Thus the note distinguishes Richardson-type order cancellation in the fixed N-series, degree-two modular gap contraction in the Landen transformation, and degree-three nonlinear error contraction in the trisection cascade.

1 Ramanujan's approximation and the eccentricity axis

Let

$$h = \left(\frac{a-b}{a+b}\right)^2.$$

The exact perimeter P of an ellipse admits the Gauss–Kummer expansion

$$P = \pi(a+b)\sum_{j=0}^{\infty}\binom{1/2}{j}^2 h^j,$$

equivalently

$$P = \pi(a+b)\left(1 + \frac{1}{4}h + \frac{1}{64}h^2 + \frac{1}{256}h^3 + \frac{25}{16384}h^4 + \cdots\right).$$

Ramanujan's second approximation reproduces this expansion exactly through order h^4 [1]. Its first discrepancy occurs at order h^5, with coefficient

$$-\frac{3}{131072}h^5.$$

Thus Ramanujan's formula is not merely a numerical fit. It is a closed expression whose Taylor expansion agrees with the first five Gauss–Kummer coefficients. This explains both its exceptional accuracy at moderate eccentricity and its fixed error floor near the degenerate limit $b \to 0$. It is an approximation along the eccentricity axis h.

2 The geometric N-series and the resolution axis

The geometric N-series construction [9] acts along a different axis. It does not replace Ramanujan's eccentricity expansion. Instead, it introduces a resolution parameter N and cancels the even-power error hierarchy

$$P_N = P + \frac{C_2}{N^2} + \frac{C_4}{N^4} + \frac{C_6}{N^6} + \cdots.$$

The corresponding acceleration operators are polynomial-resolution operators. After cancellation of the first s even-power terms, the leading remaining error is $O(N^{-2(s+1)})$. Equivalently, if the final retained order is indexed by $r = s+1$, the accelerated error is $O(N^{-2r})$. This is the classical mechanism of Richardson extrapolation and moment cancellation [7, 8].

This mechanism is fundamentally different from the arithmetic–geometric mean and Ramanujan-type rapidly convergent series for $1/\pi$, which are super-exponential in their natural iteration variable. Thus, on the ellipse, the two mechanisms are orthogonal in a precise sense: Ramanujan compresses the eccentricity structure into a closed form, while the N-series refines the geometric resolution structure by even-power cancellation.

3 The degree-two Landen bridge

The separation between the eccentricity axis and the resolution axis is not absolute. In the Gauss–Legendre arithmetic–geometric mean setting, the normalized gap lies directly on the modulus axis. Let

$$\delta_n = \frac{a_n - b_n}{a_n + b_n}$$

be the normalized gap of the arithmetic–geometric mean iteration. A direct calculation gives the local Landen contraction

$$\delta_{n+1} \sim \frac{1}{4}\delta_n^2.$$

The map $\delta_n \mapsto \delta_{n+1}$ is the descending Landen transformation of the elliptic modulus [5]. The constant $1/4$ is therefore not an arbitrary fitting constant; it is the exact local contraction coefficient of the degree-two modular transformation.

Now let

$$e_n = \pi_n - \pi, \qquad \alpha_n = -\frac{\pi_{n+1} - \pi}{\pi_{n+1} - \pi_n}$$

denote the signed Gauss–Legendre error and the ideal one-step dynamic weight.

The gap-squared error law (classical). The required error law is not new. Brent's analysis of the Gauss–Legendre algorithm gives, for the lower approximant, the doubly–exponential asymptotic

$$e_n \sim -\pi^2\, 2^{n+4}\, q^{\,2^{n+1}}, \qquad n \to \infty,$$

where q is the elliptic nome [3, 2]. The indexing here follows the present convention $e_n = \pi_n - \pi$; alternative normalizations shift n but give the same gap-squared form. The normalized AGM gap satisfies the classical relation

$$\delta_n \sim 4\, q^{\,2^n},$$

so that $\delta_n^2 \sim 16\, q^{\,2^{n+1}} = 2^4\, q^{\,2^{n+1}}$. Substituting,

$$-\pi^2\, 2^n\, \delta_n^2 \;\sim\; -\pi^2\, 2^n \cdot 2^4\, q^{\,2^{n+1}} \;=\; -\pi^2\, 2^{n+4}\, q^{\,2^{n+1}},$$

which is exactly Brent's law. Hence

$$\boxed{e_n \sim -\pi^2\, 2^n\, \delta_n^2}$$

is simply the classical Gauss–Legendre error asymptotic rewritten in the normalized gap variable, not an independent result. We use it in this form because it is the form that couples directly to the Landen contraction.

Combining the gap-squared error law with

$$\delta_{n+1} \sim \frac{1}{4}\delta_n^2$$

yields the one-eighth linkage. Since $\alpha_n = -e_{n+1}/(e_{n+1} - e_n) \sim e_{n+1}/e_n$ (because $|e_{n+1}| \ll |e_n|$), and

$$e_n \sim -\pi^2 2^n \delta_n^2, \qquad e_{n+1} \sim -\pi^2 2^{n+1} \delta_{n+1}^2,$$

we obtain

$$\alpha_n \sim \frac{e_{n+1}}{e_n} \sim 2\,\frac{\delta_{n+1}^2}{\delta_n^2} \sim 2\,\frac{(\delta_n^2/4)^2}{\delta_n^2} = \frac{1}{8}\,\delta_n^2.$$

Therefore

$$\frac{\alpha_n}{\delta_n^2} \longrightarrow \frac{1}{8} = \frac{2}{4^2}.$$

Here the numerator 2 is the Gauss–Legendre doubling factor, while the denominator 4^2 is the squared Landen contraction scale. This shows that the coefficient law detected by the dynamic error operator agrees with the classical modular contraction law at the degree-two boundary. This is the first bridge: the geometric coefficient law and the Landen modular law meet in the normalized gap variable.

Remark 1 (Consistency with the dynamic coefficient law)**.** *The one-eighth linkage is consistent with the independently established limit $\alpha_{n+1}/\alpha_n^2 \to 1/2$ for the Gauss–Legendre dynamic weights. Indeed, from $\alpha_n \sim \frac{1}{8}\delta_n^2$ and $\delta_{n+1} \sim \frac{1}{4}\delta_n^2$,*

$$\alpha_{n+1} \sim \frac{1}{8}\delta_{n+1}^2 \sim \frac{1}{8}\left(\frac{1}{4}\delta_n^2\right)^2 = \frac{1}{128}\,\delta_n^4, \qquad \alpha_n^2 \sim \frac{1}{64}\,\delta_n^4,$$

so

$$\frac{\alpha_{n+1}}{\alpha_n^2} \longrightarrow \frac{1/128}{1/64} = \frac{1}{2}.$$

Thus the linkage constant $1/8$ and the coefficient-ratio limit $1/2$ are two views of the same underlying gap dynamics.

4 Derivation of the higher-degree modular gap law

The degree-two Landen bridge raises a sharper question. Is the coefficient 1/4 only a special feature of the quadratic transformation, or is it the first member of a higher-degree modular gap hierarchy? We derive the higher-degree law directly from the small-q expansion of the modular lambda function [4].

Let q denote the elliptic nome. We use the standard convention $q = e^{i\pi\tau}$ with $\operatorname{Im}\tau > 0$; in the numerical illustrations, q is real and $0 < q < 1$. Define

$$\lambda(q) = \frac{\vartheta_2(q)^4}{\vartheta_3(q)^4}.$$

Set

$$k(q)^2 = \lambda(q), \qquad k'(q) = \sqrt{1-\lambda(q)},$$

and introduce the normalized complementary-modulus gap

$$\delta(q) = \frac{1-k'(q)}{1+k'(q)}.$$

For small q, the modular lambda function has the expansion

$$\lambda(q) = 16q - 128q^2 + 704q^3 - 3072q^4 + O(q^5).$$

Since $k'(q) = \sqrt{1-\lambda(q)}$,

$$k'(q) = 1 - 8q + 32q^2 - 96q^3 + 256q^4 + O(q^5).$$

Therefore

$$1 - k'(q) = 8q - 32q^2 + 96q^3 - 256q^4 + O(q^5),$$
$$1 + k'(q) = 2 - 8q + 32q^2 - 96q^3 + 256q^4 + O(q^5),$$

and substitution into $\delta(q) = (1-k')/(1+k')$ gives

$$\delta(q) = 4q - 16q^3 + O(q^5),$$

equivalently

$$\delta(q) = 4q\big(1 - 4q^2 + O(q^4)\big).$$

The leading relation $\delta(q) \sim 4q$ is the classical small-nome behaviour of the normalized modulus gap. Everything that follows at leading order is a corollary of this classical small-nome behaviour. The point isolated here is the first-correction coefficient and its interpretation in the present N-series/Landen comparison.

This short expansion contains both the leading law and the first correction term. For fixed $m \geq 2$,

$$\delta(q^m) = 4q^m\big(1 - 4q^{2m} + O(q^{4m})\big),$$

while

$$\delta(q)^m = 4^m q^m \big(1 - 4q^2 + O(q^4)\big)^m.$$

Hence

$$\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\,\frac{1 - 4q^{2m} + O(q^{4m})}{\big(1 - 4q^2 + O(q^4)\big)^m}.$$

Since $m \geq 2$, the numerator contributes no q^2-term. Using $(1+x)^{-m} = 1 - mx + O(x^2)$ with $x = -4q^2 + O(q^4)$,

$$\big(1 - 4q^2 + O(q^4)\big)^{-m} = 1 + 4m\,q^2 + O(q^4).$$

Therefore

$$\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\big(1 + 4m\,q^2 + O(q^4)\big).$$

We have proved the following.

Proposition 1 (Higher-degree modular gap law)**.** *Let*

$$\delta(q) = \frac{1-k'(q)}{1+k'(q)}, \quad k'(q) = \sqrt{1-\lambda(q)}, \quad \lambda(q) = \frac{\vartheta_2(q)^4}{\vartheta_3(q)^4}.$$

For every fixed integer $m \geq 2$,

$$\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\big(1 + 4m\,q^2 + O(q^4)\big), \qquad q \to 0.$$

In particular,

$$\frac{\delta(q^m)}{\delta(q)^m} \longrightarrow 4^{1-m}.$$

The leading coefficient 4^{1-m} comes from the leading term $\delta(q) \sim 4q$ and is, in that sense, classical. The first correction coefficient $4m$ comes from the next nonzero term $\delta(q) = 4q(1 - 4q^2 + O(q^4))$. The contribution here is to isolate this first-correction coefficient and interpret it in the present N-series/Landen comparison: the higher-degree gap hierarchy is governed, at first correction, by a simple linear-in-m law.

5 Numerical illustration of the higher-degree law

The proposition proves the higher-degree modular gap law for every fixed integer $m \geq 2$. We record the high-precision computations that illustrate the convergence. For fixed $m \geq 2$, define

$$R_m(q) = \frac{\delta(q^m)}{\delta(q)^m}.$$

The proposition gives $R_m(q) \to 4^{1-m}$ as $q \to 0$. The observed limiting constants for $m = 2, \ldots, 6$ are

m	4^{1-m}	observed limiting value
2	1/4	0.25
3	1/16	0.0625
4	1/64	0.015625
5	1/256	0.00390625
6	1/1024	0.0009765625

Thus the Landen constant 1/4 appears as the degree-two member of the modular gap hierarchy $\delta(q^m)/\delta(q)^m \to 4^{1-m}$. The case $m = 3$ provides a useful non-dyadic check of the formula. It is not obtained by merely repeating the displayed degree-two comparison, yet $R_3(q) \to 1/16 = 4^{1-3}$

exactly as Proposition 1 predicts. This is an illustration of the general proposition, not independent evidence for a separate modular theorem.

The first correction term is illustrated by the diagnostic

$$C_m(q) = \frac{1}{q^2}\left(\frac{R_m(q)}{4^{1-m}} - 1\right),$$

for which the proposition predicts $C_m(q) \to 4m$. The observed limiting values are

m	$4m$	observed limiting value
2	8	8.00000000000000000000000016
3	12	12.00000000000000000000000054
4	16	16.00000000000000000000000104
5	20	20.00000000000000000000000170
6	24	24.00000000000000000000000252

These values illustrate the sharper expansion proved above:

$$\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\big(1 + 4m\,q^2 + O(q^4)\big).$$

The displayed values were obtained by evaluating $C_m(q)$ at decreasing positive nomes q using high-precision arithmetic; the final displayed row corresponds to the smallest tested q. The computations are not the proof of the result; the proof follows from the theta-function expansion. The numerical values are illustrative only; the limiting constants are proved analytically in Proposition 1.

6 Guarded comparison of convergence

The preceding results strengthen the structural bridge, but they do not erase the distinction between the relevant convergence classes. The fixed-scale N-series operators act on a polynomial hierarchy in the resolution variable:

$$P_N = P + \frac{C_2}{N^2} + \frac{C_4}{N^4} + \frac{C_6}{N^6} + \cdots.$$

Their weights are selected by moment-cancellation conditions on powers of N^{-2}. No fixed finite collection of such weights turns this mechanism into a modular transformation or a super-exponential algorithm.

By contrast, the arithmetic–geometric mean and modular transformations act on the nome or modulus variable. In that setting the active gap is raised to higher powers under modular multiplication. The descending Landen transformation gives the degree-two local law

$$\delta_{n+1} \sim \frac{1}{4}\delta_n^2,$$

and Proposition 1 gives the corresponding fixed-degree small-nome hierarchy

$$\delta(q^m) \sim 4^{1-m}\delta(q)^m.$$

A third comparison point is supplied by the companion Baker 4:3 trisection cascade [11]. There the active variable is not the elliptic nome, the modulus, or the resolution scale N^{-2}, but the signed trisection error. After normalization, its local residual law has the degree-three form

$$\eta_{k+1} = -\eta_k^3 + O(\eta_k^5).$$

The correct conclusion is therefore guarded. Fixed N-series extrapolation is not itself a modular transformation, and these mechanisms do not share one convergence rate. What they share is a coefficient-selection principle: each refinement law selects distinguished weights or scalings that force its natural error variable into a higher-order or higher-power regime. In the N-series the selected weights cancel successive powers of N^{-2}; in the Landen setting the normalized gap is squared with leading coefficient $1/4$; and in the companion trisection cascade the normalized signed error is cubed.

Remark 2 (Interpretation)**.** *The phrase "structural bridge" is used deliberately. The result does not claim that geometric N-series operators are modular transformations, nor that the Baker trisection cascade is modular. It claims that the three mechanisms exhibit coefficient-selected error suppression in their own natural variables: N^{-2} for the geometric resolution hierarchy, $\delta(q)$ for the modular gap hierarchy, and the signed angular error for the* 4:3 *cascade.*

7 Conclusion

The relationship between Ramanujan, Landen, and the geometric N-series is a two-axis relationship. Ramanujan's second ellipse approximation compresses the eccentricity structure into a single exceptional closed form. The geometric N-series refines the resolution structure systematically by cancellation of an even-power hierarchy. These mechanisms are distinct on the ellipse.

They meet, however, in the modulus variable of the Gauss–Legendre arithmetic–geometric mean. There the normalized gap obeys the Landen law

$$\delta_{n+1} \sim \frac{1}{4}\delta_n^2,$$

and the classical Gauss–Legendre error asymptotic, rewritten in that gap variable, yields the one-eighth linkage

$$\frac{\alpha_n}{\delta_n^2} \longrightarrow \frac{1}{8} = \frac{2}{4^2}.$$

This is consistent with the independently proved coefficient law

$$\frac{\alpha_{n+1}}{\alpha_n^2} \longrightarrow \frac{1}{2}$$

for the dynamic Gauss–Legendre correction weights [10].

The main contribution of this note is the first-correction law in the higher-degree small-nome hierarchy. The leading relation

$$\frac{\delta(q^m)}{\delta(q)^m} \longrightarrow 4^{1-m}$$

is a direct consequence of the classical asymptotic $\delta(q) \sim 4q$, whereas the next term satisfies

$$\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\big(1 + 4m\,q^2 + O(q^4)\big).$$

Thus the degree-two Landen constant $1/4$ is the first member of a hierarchy whose first-correction coefficient is the simple linear law $4m$.

This does not identify N-series operators with modular transformations. It locates a structural resonance: each framework uses selected coefficients or scalings to suppress the active error. The companion Baker 4:3 trisection cascade [11] supplies a separate geometric instance, with a normalized cubic residual law. The three mechanisms should therefore be read as distinct but comparable:

N-series refinement is Richardson-type cancellation on the resolution axis, Landen refinement is modular contraction on the gap axis, and the trisection cascade is nonlinear geometric contraction on the angular-error axis. Whether fixed m-scale N-series operators admit a deeper modular interpretation remains open; no such identification is assumed here.

A Numerical verification data

This appendix records the numerical quantities used to illustrate Proposition 1. The computations were performed at high precision using the theta-function definition

$$\lambda(q) = \frac{\vartheta_2(q)^4}{\vartheta_3(q)^4}, \quad k'(q) = \sqrt{1-\lambda(q)}, \quad \delta(q) = \frac{1-k'(q)}{1+k'(q)}.$$

For each fixed integer $m \geq 2$, define $R_m(q) = \delta(q^m)/\delta(q)^m$. The leading law predicts $R_m(q) \to 4^{1-m}$ as $q \to 0$; the values are those of Section 5. To test the first correction, define $C_m(q) = q^{-2}\big(R_m(q)/4^{1-m} - 1\big)$, predicted to tend to $4m$; the values are likewise those of Section 5. The numerical data therefore illustrate both the leading law and the refined expansion. The computations are not used as the proof; the proof follows from the theta-function expansion in Section 4 [6].

B Classical references and positioning

The purpose of this note is not to rederive the classical theory of elliptic functions, the arithmetic–geometric mean, or Ramanujan's approximations; it identifies how the geometric N-series viewpoint sits relative to those structures. The natural background sources are: Ramanujan's approximation to the ellipse perimeter and its relation to Gauss–Kummer expansions [1]; the arithmetic–geometric mean and the Gauss–Legendre algorithm for π, together with Brent's error analysis [2, 3]; the descending Landen transformation as a degree-two modular transformation [5]; theta functions, the elliptic nome, and the modular lambda function $\lambda(q) = \vartheta_2(q)^4/\vartheta_3(q)^4$ [4, 6]; and Richardson-type extrapolation and moment cancellation as the classical analogue of the fixed-scale N-series operators [7, 8]. The related Baker papers provide the specific N-series construction [9], the independent AGM coefficient law [10], and the companion cubic trisection cascade [11].

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